

Enthusiast, Leader & Achiever Course

PHASE : (All Phase)

TARGET : PRE-MEDICAL 2020

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 05-05-2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	1	3	4	1	4	1	4	2	4	1	4	4	4	2	3	4	3	1	1	1	3	2	3	3	2	3	3	3	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	1	2	2	1	2	2	3	1	4	2	3	1	2	4	3	2	3	3	3	4	4	4	3	1	2	2	3	4	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	4	2	4	4	2	1	3	4	2	1	3	3	3	3	4	2	1	2	1	4	3	2	3	4	2	1	4	3	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	4	1	4	1	2	2	4	4	2	4	3	2	3	4	2	1	4	4	4	3	1	1	2	2	3	4	2	2	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	4	2	1	1	2	1	3	4	3	2	2	4	1	4	1	2	2	3	2	4	1	4	4	4	1	3	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	4	2	1	2	4	1	2	3	2	4	3	3	1	3	2	4	2	2	3	3	3	4	3	4	4	3	3	3	4

HINT - SHEET

2. **Ans (1)**

Heat loss = Heat gain

$$mL_v + mS_w(100 - 90) = 22S_w(90 - 20)$$

$$m[540 + 10] = 22 \times 1 \times 70$$

$$m = \frac{22 \times 70}{550} = 2.8 \text{ gm}$$

$$\text{Mass of water} = 22 + 2.8 = 24.8 \text{ gm}$$

7. **Ans (1)**

$\Delta U \rightarrow$ depends on initial and final state

$$\Delta U_1 = \Delta U_2 = U_B - U_A$$

8. **Ans (4)**

$$PT = \text{const.} \quad \dots (1)$$

$$\frac{T^n}{P^{n-1}} = \text{const.}$$

$$P^{1-n} T^n = \text{const.}$$

$$PT^{\frac{n}{1-n}} = \text{const.} \quad \dots (2)$$

By equation (1) & (2)

$$\Rightarrow \frac{n}{1-n} = 1 \Rightarrow n = \frac{1}{2}$$

$$C_{\text{poly}} = C_v + \frac{R}{1-n}$$

$$\Rightarrow 4.5R = \frac{f}{2}R + \frac{R}{1-\frac{1}{2}} \Rightarrow f = 5$$

10. **Ans (4)**

$$\frac{^\circ\text{C} - 0}{100 - 0} = \frac{W - W_0}{W_w - W_0}$$

$$\frac{C}{100} = \frac{W - 39}{239 - 39}$$

$$\frac{39}{100} = \frac{W - 39}{200}$$

$$w = (39 \times 2) + 39$$

$$= 117^\circ \text{ W}$$

12. **Ans (4)**

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$\Rightarrow \omega = \sqrt{\frac{k}{m}} \Rightarrow k = m\omega^2$$

as spring constant k remain same

$$\therefore \text{total energy } E = \frac{1}{2}kA^2 \text{ remain same}$$

15. **Ans (2)**

Time period is independent of mass of bob of pendulum.

16. **Ans (3)**

$$\text{Resultant amplitude} = \sqrt{3^2 + 4^2} = 5$$

17. Ans (4)

$$T = 2\pi \sqrt{\frac{hp}{\rho, g}}$$

Replacing h by ℓ , ρ by d and $\rho\ell$ by ρ , we get

$$T = 2\pi \sqrt{\frac{\ell d}{\rho g}}$$

18. Ans (3)

$$\text{Path difference } \Delta = \frac{\lambda}{2\pi} \times \phi$$

$$1 = \frac{\lambda}{2\pi} \times \frac{\pi}{2}$$

$$\lambda = 4 \text{ m}$$

$$V = n\lambda = 120 \times 4 = 480 \text{ m/s}$$

20. Ans (1)

$$v = \sqrt{\frac{K}{\rho}} \therefore K = v^2 \rho = 2.86 \times 10^{10} \text{ N/m}^3$$

21. Ans (1)

$$n' = n \left(\frac{v + v_t}{v - v_t} \right) = 1000 \left(\frac{330 + 220}{330 - 220} \right) \\ = 1000 \times \frac{550}{110} = 5000 \text{ Hz}$$

24. Ans (3)

As there is no approach and recession between source and observer. So observer will hear original frequency.

25. Ans (3)

Two possibilities are there

$$(i) 256 - n = 5 \text{ and } (ii) n - 256 = 5$$

when tension increased, n will increase

so $256 - n' = 2$ is possible

$$\therefore 256 - n = 5 \Rightarrow n = 256 - 5 = 251 \text{ Hz}$$

30. Ans (1)

$$\text{rms speed } v_{\text{rms}} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2}{3}}$$

$$= \sqrt{\frac{1^2 + 1^2 + 1^2}{3}} = 1 \text{ km s}^{-1}$$

$$\text{Average speed } v_{\text{av}} = \frac{v_1 + v_2 + v_3}{3}$$

$$= \frac{1 + 1 + 1}{3} = 1 \text{ km s}^{-1} = \frac{v_{\text{rms}}}{v_{\text{av}}} = \frac{1}{1} = 1$$

32. Ans (1)

$$\frac{Q}{W} = \frac{\mu C_P \Delta T}{\mu R \Delta T} \text{ (as P is constant)}$$

$$= \frac{C_P}{R} = \frac{C_P}{C_P - C_V} = \frac{C_P/C_V}{C_P/C_V - 1} = \frac{\gamma}{\gamma - 1}$$

33. Ans (2)

Coefficient of performance of refrigerator

$$\text{COP} = \frac{T_L}{T_H - T_L}$$

Where $T_L \rightarrow$ lower Temperature

& $T_H \rightarrow$ Higher Temperature

$$\text{So, } 5 = \frac{T_L}{T_H - T_L}$$

$$\Rightarrow T_H = \frac{6}{5} T_L = \frac{6}{5} (253) = 303.6 \text{ K}$$

36. Ans (2)

$$T = 2\pi \sqrt{\frac{m}{k}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{m_1}{m_2}} \Rightarrow \frac{f_1}{f_2} = \sqrt{\frac{m_2}{m_1}}$$

$$\Rightarrow m_2 = 80 \text{ kg, } m_a = 60 \text{ kg}$$

38. Ans (3)

When pendulum oscillates in air, it will lose energy continuously in overcoming resistance due to air. Therefore total mechanical energy decreases continuously with time.

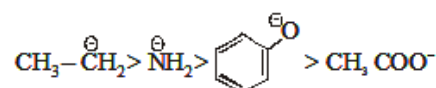
44. Ans (2)

$$n_p = \frac{v}{4L}, n_q = \frac{v}{2L}$$

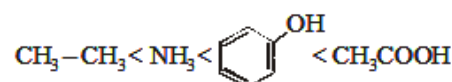
$$n_r = \frac{2v}{2L}, n_s = \frac{3v}{4L}$$

$$\therefore n_p : n_q : n_r : n_s = 1 : 2 : 4 : 3$$

54. Ans (3)



[Decreasing order of basic strength]



[Increasing order of acidic strength]

55. Ans (1)

Aromatic compound make loop-current

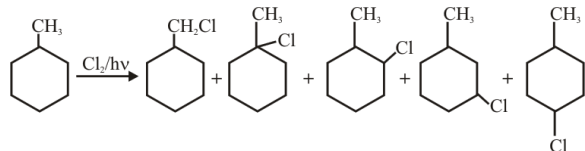
57. **Ans (2)**

Heat of hydrogenation $\propto \frac{1}{\text{stability of Alkene}}$

60. **Ans (2)**

due to -M effect of $-\text{NO}_2$ group at ortho position

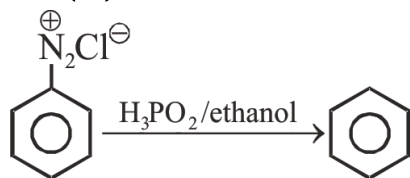
63. **Ans (2)**



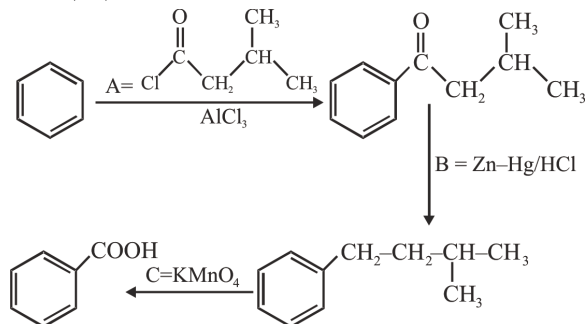
69. **Ans (4)**

Saturated and Benzene can't show test of unsaturation.

70. **Ans (2)**



74. **Ans (3)**



81. **Ans (4)**

Stability of alkene $\propto$ no. of alpha Hydrogen atom

90. **Ans (4)**

For keto to enol $\rightarrow$ one acidic H must be in conjugation with keto grp.

91. **Ans (4)**

NCERT-XI, Pg. # 331

93. **Ans (1)**

NCERT XIth Pg.# 332 Para-22.2.2

97. **Ans (2)**

NCERT (XI) Pg. # 338

101. **Ans (4)**

NCERT (E) Pg. # 317

103. **Ans (2)**

NCERT (XIth) Pg. # 321, Para - 1

109. **Ans (4)**

NCERT Pg.# 323

112. **Ans (1)**

NCERT XIth Pg.# 325

115. **Ans (2)**

NCERT-Page No.102

127. **Ans (2)**

NCERT Pg. # 312

128. **Ans (1)**

NCERT-XI Pg # 281

133. **Ans (2)**

NCERT (XI) Pg # 283

145. **Ans (4)**

NCERT XI, Pg. No. # 263

147. **Ans (1)**

NCERT (XIth) Pg. # 294

152. **Ans (4)**

NCERT-XI, Pg. # 298

154. **Ans (1)**

NCERT Pg # 291

159. **Ans (3)**

NCERT XI- Pg. # 274

161. **Ans (4)**

NCERT (XI) Pg # 276

165. **Ans (3)**

NCERT (XIth) Pg. # 115

166. **Ans (2)**

NCERT-XI, Pg#114

167. **Ans (4)**

NCERT Pg. # 50,53

168. **Ans (2)**

NCERT-XI - Page No.-46, "Levels of organization"

176. **Ans (4)**

NCERT (XI) (E) Pg. # 56, 57

178. **Ans (3)**

NCERT (XI) Pg. # 60